Compulsory Exercise

Topics in Life Insurance (Liv2), Block 3, 2026

Mogens Bladt & Jamaal Ahmad

Formalities

- The exercise is due on Sunday 22/3/2026 at 23:59
- It must be handed in groups of up to 3 via Absalon
- It must be approved (in the first attempt) to participate in the exam

Exercise

In this exercise, we consider a traditional Danish life-cycle unit-link product design. We examine a 30-year old today in the survival model depicted in Figure 1.

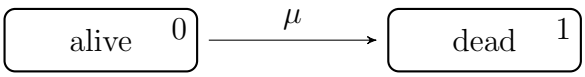

Figure 1: The survival model with mortality rate μ.

The product design works as follows. During active years on the labour market, that is, until retirement at time m, the insured pays a pension contribution as a percentage $\theta \in [0,1]$ of the labour income $L(t)$ at time t into the unit-link account, i.e.

$$b^0(t,x) = a^0(t,x) = -\theta L(t) \qquad \text{for } t \in [30, m).$$

In addition to this, a term life insurance payable upon death at time $t < m$ is bought. The size is equal to a factor $K \geq 0$ of the labour income, i.e.

$$b^{01}(t,x) = a^{01}(t,x) = KL(t), \qquad \text{for } t \in [30, m).$$

The unit-link account X until retirement thus develops according to, on the event of survival (i.e. disregarding the $\mathrm{d}N$-term),

$$\begin{aligned} \mathrm{d}X(t) &= (\pi(t, X(t))\alpha + (1 - \pi(t, X(t)))r)X(t)\,\mathrm{d}t \\ &\quad + \sigma\pi(t, X(t))X(t)\,\mathrm{d}W(t) \\ &\quad + \theta L(t)\,\mathrm{d}t - \mu(t)(KL(t) - X(t))\,\mathrm{d}t, \\ X(30) &= 0. \end{aligned} \tag{1}$$

From time m, the savings account X is continously used to pay out pension benefits in the form of a life annuity. These are bought on a pre-specified calculation basis ('udbetalingsgrundlag') given by $(\widetilde{r}, \widetilde{\mu})$, so that

$$b^0(t,x) = a^0(t,x) = \frac{x}{\widetilde{V}_0(t)}, \qquad \text{for } t \in [m, \infty),$$

where $\tilde{V}_0$ is the reserve of a life annuity in the alive state valuated on the basis $(\widetilde{r}, \widetilde{\mu})$:

$$\widetilde{V}_0(t) = \int_t^\infty e^{-\int_t^s \widetilde{r}(v)+\widetilde{\mu}(v)\,\mathrm{d}v}\,\mathrm{d}s.$$

The unit-link account X after retirement thus develops according to, on the event of survival,

$$\begin{aligned} \mathrm{d}X(t) &= (\pi(t,X(t))\alpha + (1-\pi(t,X(t)))r)X(t)\,\mathrm{d}t \\ &\quad + \sigma\pi(t,X(t))X(t)\,\mathrm{d}W(t) \\ &\quad - \frac{X(t)}{V^\star(t)}\,\mathrm{d}t - \mu(t)(-X(t))\,\mathrm{d}t. \end{aligned} \tag{2}$$

The idea behind the life-cycle products is that the insured, to some extent, participates actively in the design of the investment strategy, with a common feature that the proportion of risky assets gets lowered as the insured approaches retirement at time m. We assume the following structure:

$$\pi(t,x) = \begin{cases} 1 & \text{for } t < 55 \\ -\frac{1}{2(m-55)}t + \frac{2m-55}{2(m-55)} & \text{for } t \in [55, m], \\ \frac{1}{2} & \text{for } t > m. \end{cases}$$

This is inspired by actual profiles seen in Danish practice, cf. e.g. Appendix B below. The parameters to be used in the exercise are shown in Table 1.

Parameter	r	α	σ	m	θ	K	$L(t)$
Value	0.02	0.04	0.13	70	0.12	4	$600000 \cdot e^{0.018(t-30)}$

Table 1: Model parameters for the exercise.

Parameter	$\widetilde{r}(t)$	$\widetilde{\mu}(t)$
Value	$\pi(t)\alpha + (1-\pi(t))r$	$\mu(t)$

Table 2: Calculation basis for the life annuity

Furthermore, the mortality rate μ is obtained from the file **mortality.csv** available on Absalon. It is based on benchmark mortalities published by the Danish Financial Supervisory Authorities in 2025. Throughout the exercise, you can use a suitable interpolation method on this to create a continuous function, e.g. using **approxfun** in **R**.

Lastly, the calculation basis is given in Table 2.

(*i*) Calculate the reserve $\widetilde{V}_0$ via a backward Thiele differential equation starting from a maximal age of 111.

(*ii*) Simulate a single path of the account X until age 111 using an Euler scheme with step length $h = 1/10000$ on (1) and (2) (See Appendix A for details).

(*iii*) Plot your simulation of the account and comment on the result. How much savings do you have at retirement?

(*iv*) Argue that the market reserve is equal to the account value, i.e. $V^0(t, x) = x$.

Hint: Examine (your simulation of) $X(111)$.

(*v*) Let $Y(t) = b^0(t, X(t))$, for $t \geq m$, be the process of the life annuity benefits. Show that it is a (generalised) Geometric Brownian motion with dynamics

$$\mathrm{d}Y(t) = (\pi(t)\alpha + (1 - \pi(t))r - \widetilde{r}(t) + \mu(t) - \widetilde{\mu}(t))\, Y(t)\, \mathrm{d}t + \sigma\pi(t)Y(t)\, \mathrm{d}W(t),$$

$$Y(m) = \frac{X(m)}{\widetilde{V}(m)}.$$

How can we relate the calculation basis $(\widetilde{r}, \widetilde{\mu})$ to the payout profile the insured gets?

(*vi*) Use the dynamics from (*v*) to simulate $N = 1000$ paths of the life annuity benefits using an Euler scheme with step length $h = 1/100$. Use $X(m)$ from (*ii*) in the initial value.

(*vii*) Based on your simulations, plot the expected life annuity benefits through time given $X(m)$ from (*ii*). Comment on the payout profile you have obtained.

Appendix

A - Simulation from SDEs

Suppose we have a process $A = \{A(t)\}_{t \geq 0}$ that satisfies a stochastic differential equation (SDE) on the form

$$\mathrm{d}A(t) = f(t, A(t))\, \mathrm{d}t + g(t, A(t))\, \mathrm{d}W(t), \qquad A(0) = A_0$$

where f and g are suitably regular, deterministic functions, and W is a standard Brownian motion.

The easiest and most common way to simulate paths of the process A on some time interval $[0, T]$ using its SDE is to adopt the Euler method, described as follows. Construct a grid on $[0, T]$ as:

$$0 = t_0 < t_1 < \cdots < t_M = T,$$

where M is the number of time points on the grid. Then, for $i = 0, \ldots, M-1$, one moves a time forward:

$$A(t_{i+1}) = A(t_i) + f(t_i, A(t_i))h + g(t_i, A(t_i))\sqrt{h}Z_i,$$

where $Z_0, \ldots, Z_{M-1}$ are i.i.d standard normal distributed random variables.

Usually, one adopts an equdistant grid with some pre-specified length $h > 0$ such that $t_{i+1} - t_i = h$ for all i.

B - Investment strategies

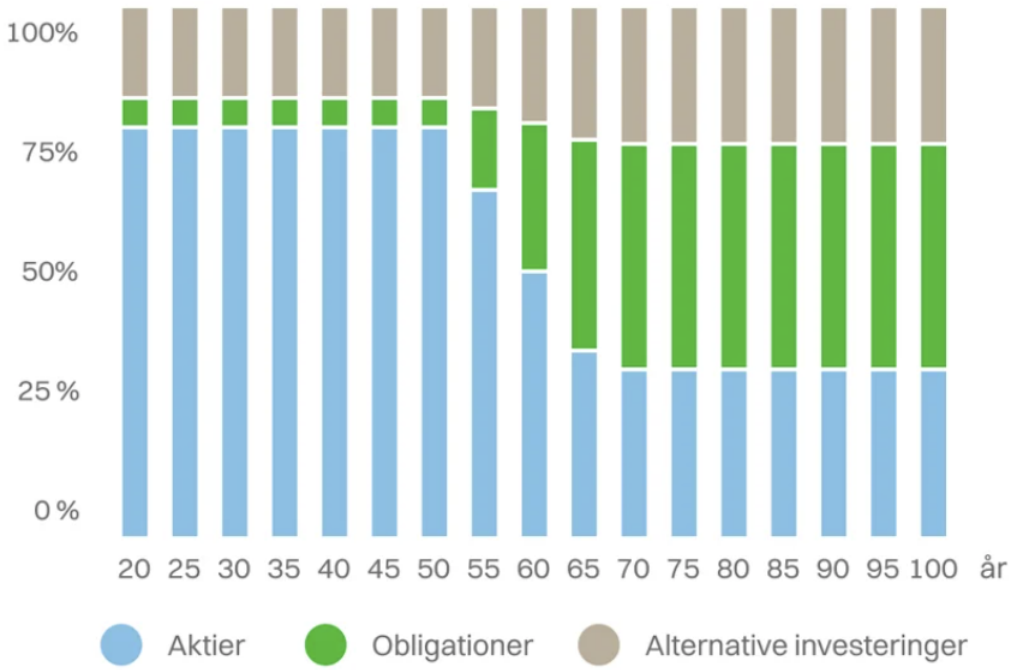

Figure 2: Example of a life-cycle investment profile in a Danish life insurance company